Design Computing Studio 2

Interaction Design

Assessment 2: Project

Due Dates: Starting from 6th Sept 2024 1:00 PM – 14th Nov 2024 15:00.
Weighting: 60% [Undergrad] /50% [Postgrad]
Team Work

Description:

Students forming teams and working on their projects is at the core of Studio-2 Interaction Design. As a part of this assessment, student teams will identify a problem space and develop a software solution, using various Interaction Design methods and techniques. Teams will follow a complete design cycle of exploring and understanding a problem, eliciting user requirements, designing working software prototypes and testing/evaluating them in real-world situations.

Teams will conduct two iterations of their prototypes, and showcase their design prototypes and processes in week 7 (first iteration) and in week 13 (second iteration). A detailed project report of the complete design and development process is expected at the end of the course.

This team-based assessment will be divided into three smaller-assessment milestones:

1. Design Exhibit/Presentation of Iteration 1 (week 7): 15% [UG/PG]
2. Design Exhibit of Iteration 2 (week 13): 25% [UG/PG]
3. Project Report (14th Nov 2024): 20% [UG] / 10% [PG]

Learning Objectives:

1 Analyse and apply concepts of Interaction Design in software design and development.
2 Create a successful and multi-disciplinary team culture, and reflectively apply methods and techniques for effective collaborative teamwork, including the use of digital tools and platforms, resolving conflicts where they arise.
3 Apply user research, prototyping and evaluation techniques to a specific problem area.
4 Effectively communicate the outcomes of the design project using various modalities and forms.

Deliverable:

The deliverables of this assessment item are divided in three milestones.

1. **Milestone 1 - Design Exhibit of Iteration 1.** This deliverable is expected to be a work-in-progress activity, where a prototype will be exhibited during a regular studio session. It is expected that prototypes developed in this milestone will have some level of 'interactivity'. There is no expectation of any user evaluations/testing in this milestone. This assessment will take place during week 7's regular studio hours.

2. **Milestone 2 - Design Exhibit of Iteration 2.** This deliverable should present a high-fidelity prototype, emphasising detailed aesthetics, interactions, and a polished look-and-feel. The prototype should integrate findings and insights from user engagement, testing, and evaluation. This assessment is scheduled during the regular studio hours of week 13.
3. **Milestone 3 - Project Report.** This deliverable is expected to be a ~3000 words project report that should capture all aspects of the project.

Further Information:

Studio 2 students will engage in this assessment item throughout the course. Students will form a team of 4-6 members in week 2 and will be able to define their own projects based on the Project Themes provided on the Blackboard site. It is important to note that the course coordinator Dr Dhaval Vyas recommends to have a team of 6 to maximise the efforts required for this project. However, students can choose to have a team of 4 or 5 if they want. It will be the students' responsibility to ensure that their projects meet all the requirements for this assessment. There will be no special considerations given to a team of less than 6 members.

The Process:
Team should follow the following steps:

Weeks	Activities	Outcomes
Week 1-2	Form a Team and Define a Project	• A team of 6 students • A project description (no software solutions or apps at this stage)
Week 3-5	Plan a study to investigate the problem further • Develop a suite of methods (interviews, observations, etc.) to find answer to the project question • Engage with potential users and collect data • Each team member should engage with at least one participant and conduct a specific method. I.e., in a 6-member team, all 6 students need to have engaged with at least 1 participant • It is expected that you will have 6-10 participants in your study • Add ethical insights and principles in your approach	• A set of interview transcripts or other forms of evidence of studies conducted • A design concept based on data analysis
Week 6-7	Prototype for Iteration 1 • Your prototype should indicate main functionalities • It should have a level of dynamic behaviour in your prototypes	• A prototype web or hardware application • Evidence of design process undertaken (design sketches, storyboards, and other forms of prototypes)
Week 8-11	Run evaluation / user enactment / assessment on your prototypes • Select appropriate evaluation methods • Engage with potential users and deploy your prototypes in real-world contexts • Each team member should engage with at least one participant and conduct evaluation	• A functional prototype that is deployed in real-world settings • A set of improvement list based on user feedback

	• It is expected that you will have 4-6 participants in your evaluation study	
Week 10-12	Prototype for Iteration 2 • Your prototype should be close to a final product • You should incorporate evaluation findings into this iteration	• A functional prototype with realistic look-and-feel, which includes user feedback

Milestone 1 - Design Exhibit of Iteration 1

Due Date: 6th September 2024, Design Exhibition regular studio hours
Weighting: 15%
Team Work

Description:

Teams will present their exhibit their prototype iteration 1. These exhibits will be opportunities to give and receive feedback on progress and peer critique.

Deliverable:

- Teams will create an "exhibition booth" around their tables showcasing all the evidence of their work thus far.
- Teams will present a 5-minute pitch
- All team members must be present and actively contribute to the exhibit.

Assessment Criteria

Criterial	7 – High Distinction	6 - Distinction	5 – Credit	4 – Borderline Pass	3/2/1 - Fail
Overall Presentation 20%	Excellent visual and verbal communication Answered all the questions correctly and effectively	Good visual and verbal communication Most questions were answered correctly and effectively	Visual and verbal communication need improvement Most questions were answered correctly and effectively	Poor visual and verbal communication Very defensive in answering questions and poor overall engagement	No or poor attempt of any form of communication
Connection with Field Data & Quality of Prototype 40%	Excellent connection made between user insights and the prototype Excellent show of prototype demo – functionalities and user interactions are clear "Dynamic" features of the prototype are clear	Good connection made between user insights and the prototype Prototype functionalities and user interactions are clear "Dynamic" features of the prototype are unclear	Unclear connection made between user insights and the prototype Functionalities and user interactions are clear No "Dynamic" features in the prototype	Poor or no connection between user insights and the prototype Unclear / No prototype functionalities No "Dynamic" features in the prototype	No or poor attempt of making connection with user insights Prototype non-existent

Creativity 40%	The design showcases exceptionally creative and original ideas that push boundaries and challenge norms.	The design incorporates creative elements that set it apart from common approaches, demonstrating originality.	The design shows some degree of creativity, but it might also contain elements that are somewhat common or predictable.	The design lacks significant creative thought and relies heavily on existing ideas or trends.	The design lacks any semblance of originality and relies entirely on conventional concepts.

Dynamic (in the context of this assessment): Refers to the interactive elements of the prototype, whether software or hardware, that respond to user input or environmental conditions. For software prototypes, dynamic elements may include areas where users can input, manipulate, or interact with information, resulting in a visible or functional change within the prototype. In the context of hardware or IoT devices, dynamic refers to components or features that react to physical interactions such as touch, motion, or environmental changes, producing a tangible result or feedback. This is in contrast to static elements or components, which remain unchangeable and non-interactive regardless of user interactions or environmental conditions. Examples of dynamic features might include touch sensors on a device that light up upon contact, motion sensors that trigger specific actions, or software buttons leading to another screen.

Milestone 2 - Design Exhibit of Iteration 2

Due Date: 25th October 2024, Design Exhibition regular studio hours
Weighting: 25%
Team Work

Description:

Teams will present their exhibit their prototype iteration 2. These exhibits will be opportunities to give and receive feedback on progress and peer critique.

Deliverable:

- Teams will create an "exhibition booth" around their tables showcasing their final prototype.
- An A2 size colour poster to describe their design process.
- One-page design flyer that can be taken away by a visitor
- Exhibit should have all the evidence of their work.
- All team members must be present and actively contribute to the exhibit.

Assessment Criteria

Criterial	7 – High Distinction	6 - Distinction	5 – Credit	4 – Borderline Pass	3/2/1 - Fail
Overall Presentation 20%	Excellent visual and verbal communication Answered all the questions correctly and effectively	Good visual and verbal communication Most questions were answered correctly and effectively	Visual and verbal communication need improvement Most questions were answered correctly and effectively	Poor visual and verbal communication Very defensive in answering questions and poor overall engagement	No or poor attempt of any form of communication
Connection with Field Data & Quality of Prototype 40%	Excellent connection made between user insights and the prototype Excellent show of prototype demo – functionalities and user interactions are clear	Good connection made between user insights and the prototype Prototype functionalities and user interactions are clear	Unclear connection made between user insights and the prototype Functionalities and user interactions are clear	Poor or no connection between user insights and the prototype Unclear / No prototype functionalities	No or poor attempt of making connection with user insights Prototype non-existent

Design Process 40%	Excellent poster and design flyer; makes the exhibit and overall project ideas clear	Good poster and design flyer; makes the exhibit and overall project ideas clear	Poster and design flyer are less engaging but the overall project ideas are clear	Poor poster and design flyer; overall unclarity around project ideas	No or poor attempt of making any kind of visuals
	Ample amount of evidence of design activities through multiple low and high-fidelity prototypes	Evidence of design activities through multiple low and high-fidelity prototypes	Little evidence of design activities through very few low and high-fidelity prototypes	Little evidence of design activities through very few low and high-fidelity prototypes	Little or no evidence of design activities; prototypes non existent
	Excellent improvements in this iteration through insights from user evaluation	Clear evidence of applying evaluation findings in this iteration	User evaluation is clear but improvements on this prototype are unclear	No attempt on conducting any type of evaluations	No attempt on conducting any type of evaluations
	Exceptionally creative exhibit that showcases a high level of originality and innovative design elements	Exhibit demonstrates moderate creativity and includes a few unique and innovative design choices.	Exhibit demonstrates moderate creativity and includes a few unique and innovative design choices	Some original elements present, but overall creativity is limited.	Exhibit lacks originality and creativity, relying heavily on common design elements.

Milestone 3 – Project Report

Due Date: 14th November 2024, 3pm via the Blackboard site
Weighting: 20% [UG] / 10% [PG]
Team Work

Deliverable:

A report on a prescribed format (~3000 words). Attach user research material and data as appendices (no word/page limit). You must acknowledge if you use generative AI tools (e.g. ChatGPT), along with a complete list of prompts, in English.

The final report should have the following contents:

a. Main report + Appendices, in an ACM SIGCHI format (available on the Blackboard site). Main report should be around ~3000 words; appendices have no word or page limit. Use APA referencing style.
b. Academic Integrity Statement (available on the Blackboard site) must be signed by all the team members and added as the first page of your report.
c. Attach evidence of your work, i.e., user data such as interview transcripts, photos, and other forms of evidence added in appendices.
d. Attach signed Ethics Forms (available on the Blackboard site) in the appendices of the report.
e. The main report should be structured as following

Title and Student Names	• Give a meaningful title to your report and add you full names, student IDs and UQ email addresses
Abstract	• Give an overall highlight and summary of your report
Introduction	• Provide an overview of the overall premise. Introduction should clearly express the purpose, objective and scope of the work. • Describe problem domain, aims and objectives, methodology, and summary of findings
Design Process	• Describe your design approach including user research, prototyping, and evaluation; focus on the outcomes of these methods • Provide justification on selecting specific methods
Discussion & Conclusion	• Discuss your findings and justify significance of your results • Draw your report together in a short conclusion.
Appendices	• Show the evidence of your study e.g. interview transcripts, photos from the field, other material, etc.

Assessment Criteria:

Report Content	7 – High Distinction	6 – Distinction	5 – Credit	4 – Borderline Pass	3 or Less – Fail
Writing and Presentation	Excellent logical structure, physical layout and appropriate attention to detail. The work is	Very good logical structure, physical layout and attention to detail. The work is presented in an	Good structure, physical layout. Some minor inaccuracies in presentation of work.	Acceptable structure and physical layout; Some inaccuracies or lack of	Structure and physical layout very poor; presentation of work is inaccurate and incomplete.

10%	presented in an accurate, concise and coherent fashion. No spelling mistakes or grammar errors. Appropriate referencing to a correctly formatted bibliography. Appropriately acknowledges the work of others.	accurate and coherent fashion. Minimal spelling mistakes or grammar errors. Appropriate referencing to a correctly formatted bibliography. Appropriately acknowledges the work of others.	Neat, occasional spelling mistakes or grammar errors. Few errors in referencing. Appropriately acknowledges the work of others.	detail in presentation of work. Neat, some spelling mistakes or grammar errors. Some errors in referencing or bibliography formatting. Appropriately acknowledges the work of others.	Numerous spelling mistakes or grammar errors. Several errors in referencing or bibliography formatting. Many failures to appropriately acknowledge the work of others
Abstract and Introduction 10%	Excellent, clear definition of a substantial and significant project area. Introduction provides an excellent overview of the design project. The abstract accurately yet concisely captures the topic, methods and outcomes.	Very good definition of problem area. Introduction clearly gives a big picture of the design project. The abstract accurately captures the topic, methods and outcomes.	Good definition of problem area. Introduction clearly gives a big picture of the design project. The abstract captures the topic and outcomes.	Satisfactory definition of the topic The overall purpose and a big picture of the design project is not well articulated. The abstract satisfactorily captures the topic and outcomes	Topic and scope of the project are very unclear. The introduction and abstract do not summarise the design project and its outcomes.
Design process 50%	Overall design approach and methods are systematically applied. Your user research highlights novel and insightful findings around the given topic.	A clearly set out design approach and methods systematically applied. Your user research findings are clearly described and provide logical insights.	Some inaccuracies in the application of design methods. Your user research findings are described comprehensively.	The design approach is reasonably systematic. Your user research findings lacked depth and did not have believability.	The design approach is haphazard, and has no logical basis. User research findings are very superficial and do not add any new knowledge to

	The description of your prototyping and evaluation process is excellent and has a strong connection with your user research findings The overall execution indicates a substantial design effort. The appendices show ample evidence about the quality of your work.	The description of your prototyping and evaluation process is excellent and has a strong connection with your user research findings The overall execution shows a good effort. The appendices show reasonable evidence about the quality of your work.	Your prototyping and evaluation process needed a bit more clarity; it did have a good connection with your user research findings The overall execution shows a good effort. The appendices show weak evidence about the quality of your work.	Your prototyping and evaluation process are weakly described and connection to your user research findings are not strong enough. The overall execution shows insufficient effort. The appendices show minimal evidence about your work.	the area being designed for. No or little indication of prototyping and evaluation The overall execution shows insufficient effort. No appendices are attached.
Discussion & Conclusion 30%	The analysis of the work conducted highlights your comprehension and shows insight into the significance of the results. The project concludes with a clear concise summary of the outcomes – in context of specified topic.	The analysis of the work conducted shows some insight into the significance of the results. The project concludes with a clear summary of the outcomes.	The analysis of the work conducted shows comprehension of the work but limited insight into the significance of results. The project concludes with a summary of the outcomes.	The analysis of the work conducted demonstrates some comprehension. The project concludes with a high-level summary of outcomes.	Poor review of the work and no real comprehension demonstrated. Conclusion fails to explain what was achieved.